TSAR Project Part 1 – Exploratory Data Analysis

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```r
# # Random sample your group's data set from the original LCdata data set
# LCdata <- fread("LCdata.csv")
# set.seed(1)
# myLCdata <- LCdata %>% sample(5) %>% slice_sample(prop = .5)
# fwrite(myLCdata, file = "myLCdata.csv")
myLCdata <- fread("myLCdata.csv")
```

1 Task 1: Business Understanding

```r
glimpse(myLCdata)
```

```
Rows: 44,369
Columns: 5
$ annual_inc     <dbl> 60000, 18576, 40000, 35000, 125000, 144000, 205000, 118~
$ open_acc       <int> 6, 7, 15, 3, 12, 8, 9, 18, 12, 3, 11, 10, 15, 5, 13, 10~
$ purpose        <chr> "credit_card", "debt_consolidation", "small_business", ~
$ home_ownership <chr> "OWN", "RENT", "RENT", "OWN", "RENT", "MORTGAGE", "MORT~
$ inq_last_6mths <int> 0, 0, 2, 1, 0, 2, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 3, 0~
```

```r
str(myLCdata)
```

```
Classes 'data.table' and 'data.frame':  44369 obs. of  5 variables:
 $ annual_inc    : num  60000 18576 40000 35000 125000 ...
 $ open_acc      : int  6 7 15 3 12 8 9 18 12 3 ...
 $ purpose       : chr  "credit_card" "debt_consolidation" "small_business" "home_improvemen
```

```
 $ home_ownership: chr  "OWN" "RENT" "RENT" "OWN" ...
 $ inq_last_6mths: int  0 0 2 1 0 2 0 0 1 0 ...
 - attr(*, ".internal.selfref")=<externalptr>
```

1.1 Variable Overview and Business Interpretation

The dataset `myLCdata` contains 44,369 observations and 5 variables. Based on the output of `glimpse()` and `str()`, the variables can be categorized as follows:

1.1.1 Variable Description

- **annual_inc (numeric)**
  The self-reported annual income provided by the borrower during registration. This variable is continuous and measures repayment capacity.
- **open_acc (integer)**
  The number of open credit lines in the borrower's credit file. This reflects the borrower's existing credit exposure.
- **purpose (character)**
  A category provided by the borrower for the loan request. (e.g., credit card refinancing, debt consolidation, small business, home improvement). This is a categorical variable.
- **home_ownership (character)**
  The home ownership status provided by the borrower during registration.(e.g., RENT, OWN, MORTGAGE, OTHER).
- **inq_last_6mths (integer)**
  The number of inquiries in past 6 months (excluding auto and mortgage inquiries). This is an indicator of recent credit-seeking behavior and potential financial stress.

1.2 Business Context in Peer-to-Peer Lending

<what to write here??????????>

2 Task 2: Data Understanding Based on Summary Statistics

2.1 Summary Statistics

```r
summary(myLCdata)
```

```
   annual_inc         open_acc       purpose           home_ownership
 Min.   :   4000   Min.   : 1.00   Length:44369       Length:44369
 1st Qu.:  45649   1st Qu.: 8.00   Class :character   Class :character
 Median :  65000   Median :11.00   Mode  :character   Mode  :character
 Mean   :  75269   Mean   :11.57
 3rd Qu.:  90000   3rd Qu.:14.00
 Max.   :6000000   Max.   :76.00
                   NA's   :2
 inq_last_6mths
 Min.   : 0.0000
 1st Qu.: 0.0000
 Median : 0.0000
 Mean   : 0.6966
 3rd Qu.: 1.0000
 Max.   :31.0000
 NA's   :2
```

```r
# install skimr if necessary
if (!require("skimr")) install.packages("skimr")
library(skimr)

skim(myLCdata)
```

Table 1: Data summary

Name	myLCdata
Number of rows	44369
Number of columns	5
Key	NULL
Column type frequency:	
character	2
numeric	3

Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
purpose	0	1	3	18	0	14	0
home_ownership	0	1	3	8	0	6	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
annual_inc	0	1	75269.01	60410.50	4000	45649	65000	90000	6000000	
open_acc	2	1	11.57	5.38	1	8	11	14	76	
inq_last_6mths	2	1	0.70	1.01	0	0	0	1	31	

2.2 Interpretation and Domain Context

2.2.1 1. Missing Values (NAs) ——

- Are missing values present?
  - Categorical values: No missing values
  - Numerical values:
    - Missing values: 2 tables open_acc, 2 inq_last_6mths
    - No missing values: annual_inc
- Is this plausible from a business perspective?
- Would this require data preparation?
  - Yes, but can be handled by removing the data objects. The risk of distorting of the data is low due to small number of data objects.

Discussion:

With such a low number missing values and the total number of data objects of 44369 it the easiest way to handle the missing values is to remove the data objects with missing values. imputation methods require pre calculations which in this case are not worth to do.

2.2.2 2. Numerical Variables

- Mean / Median:
  - The difference in mean & median seems high in the table inq_last_6mths given the distribution of the 1st Qu and 3rd Qu. This can be explained by the Max which 31 and also an outlier.
- Minimum:
  - The minimum seems reasonable in all the tables
- Maximum:
  - The maximum in annual_inc and inq_last_6mths seem to be outlier. Maybe these data objects can be removed to reduce the distortion in this tables.
- Standard deviation:
  - The std. dev. in the annual_inc table seems too high with a value of 60410. The reason could lay in the outlier of the max which is 6 mio.
- Is the distribution symmetric or skewed?
  - symmetric:
    - open_acc
  - right skewed:
    - annual_inc, inq_last_6mths
- Are there potential outliers?
  - The min of 4000 and max of 6 mio seems to be outliers in the table annual_inc
  - The max of 31 in inq_last_6mths seems to be also an outlier
- Do the extreme values seem realistic in the Lending Club context?
  - The outlier of 6 mio in the table annual_inc does not the be realistic in the lending club context. You should expect a person having an income of 6 mio does need a loan from a platform. The other outliers seem to be realistic in this context.

Statistical Interpretation: > Write here.

Domain Interpretation: > Write here.

2.2.3 3. Categorical Variables

For each categorical variable:

- Number of categories:
  - purpose: 14
  - home_ownership: 6
- Most frequent category:
  - is not readable from the summary or skim table
- Least frequent category:
  - is not readable from the summary or skim table
- Are categories balanced or highly unbalanced?
  - is not readable from the summary or skim table

Statistical Interpretation: > Write here.

Domain Interpretation: > Write here.

2.2.4 4. Key Insights from Summary Statistics

- Most important statistical findings:
- Potential data quality issues:
- First hypotheses about borrower behavior:

---

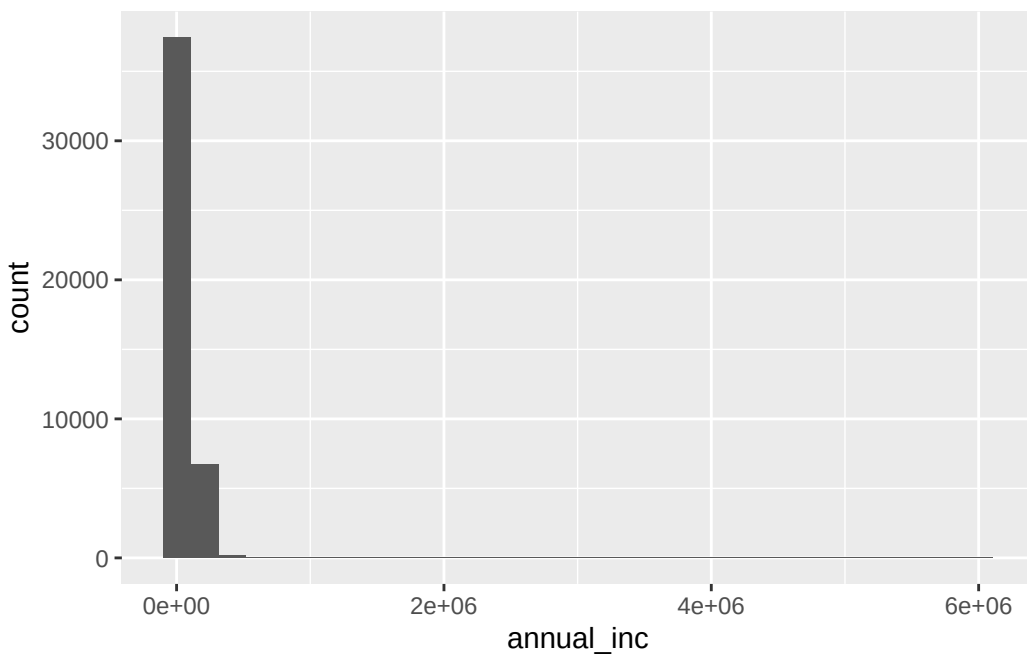
3 Task 3: Univariate Exploratory Data Analysis (EDA) with ggplot2

3.1 Variable 1: annual_inc

3.1.1 Visualization

```
# Shows distribution of income

ggplot(myLCdata, aes(x = annual_inc)) +
  geom_histogram(bins = 30)
```

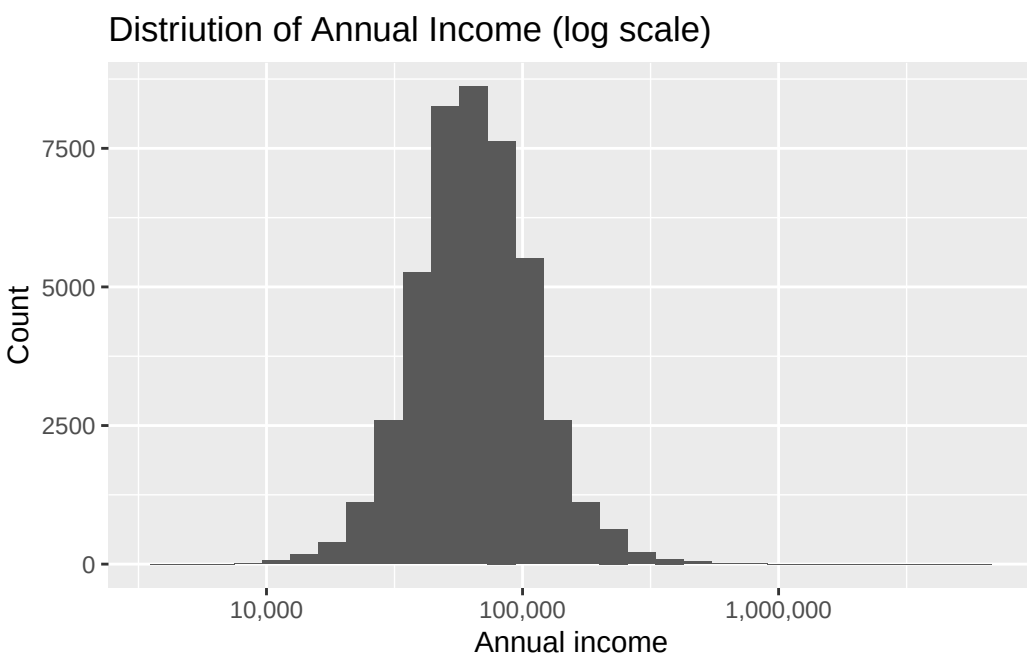


3.1.2 Interpretation:

The annual income variable is extremly right skewed due the small number of extremly high income (e.g. 6 mio income which we detected in task 2). Therefor we need to remove this extreme value:

```
library(scales)
ggplot(myLCdata, aes(x = annual_inc)) +
  geom_histogram(bins = 30) +
```

```
  scale_x_log10(labels = label_comma()) +
  labs(
    title = "Distriution of Annual Income (log scale)",
    x = "Annual income",
    y = "Count"
  )
```

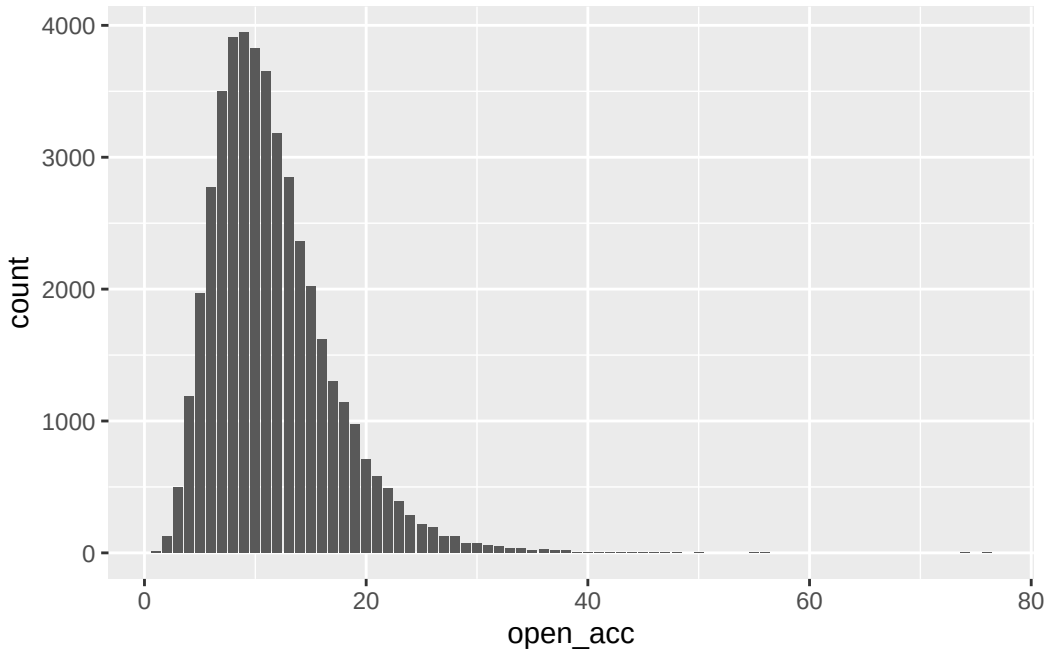


3.2 Variable 2: `open_acc`

3.2.1 Visualization

```
ggplot(myLCdata, aes(x = open_acc))+
  geom_bar()
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_count()`).
```

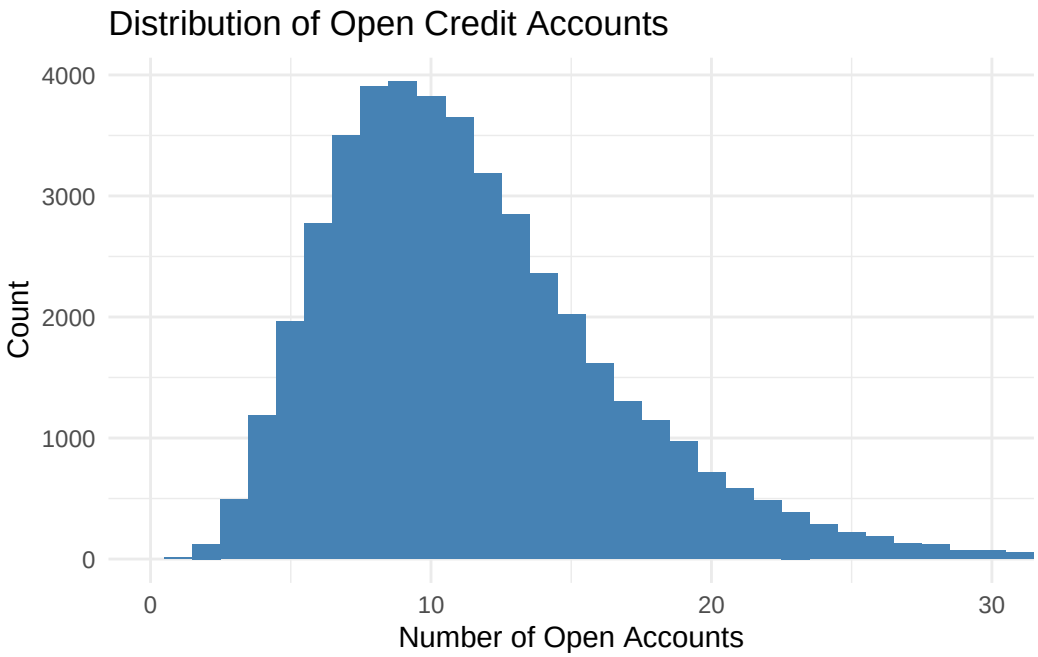


3.2.2 Interpretation

This variable is also right skewed. Again we have seen in task 2 that the highest data point is an outlier. but not as clearly as here in this visualization. Most borrowers have between 5 and 15 (Task 2) open accounts, while a small number of borrowers have significantly more accounts. We have then changed it to histogram and excluded the outliers:

```
ggplot(myLCdata, aes(x = open_acc)) +
  geom_histogram(binwidth = 1, fill = "steelblue") +
  coord_cartesian(xlim = c(0, 30)) +
  labs(
    title = "Distribution of Open Credit Accounts",
    x = "Number of Open Accounts",
    y = "Count"
  ) +
  theme_minimal()
```

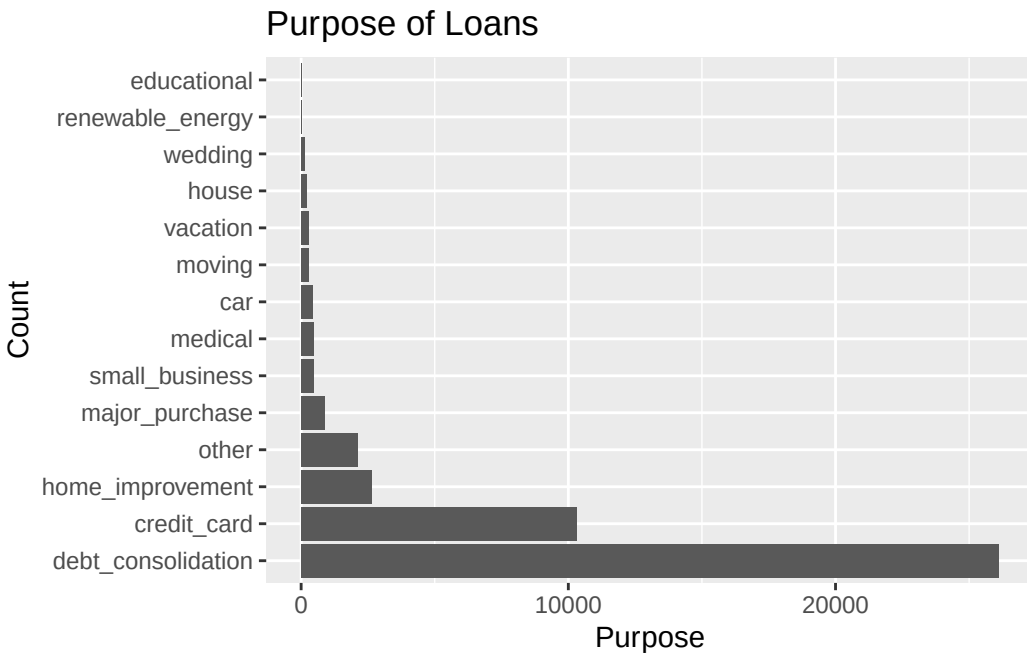
```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).
```



3.3 Variable 3: `purpose`

3.3.1 Visualization

```
ggplot(myLCdata, aes(x = forcats::fct_infreq(purpose))) +
  geom_bar() +
  coord_flip() +
  labs(
    title = "Purpose of Loans",
    x = "Count",
    y = "Purpose"
  )
```



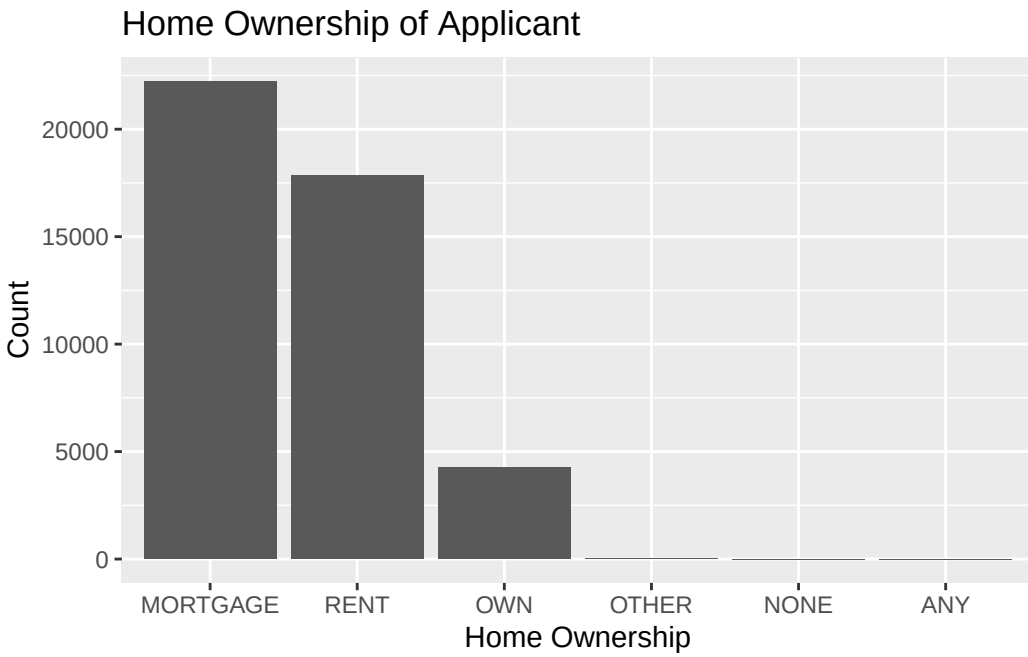
3.3.2 Interpretation

3.4 > In this chart we see clearly see that the purpose for most of the loans debt consolidation is. WE have read in the description of the task that the this site offers more competitive loans because it is a peer to peer credit marketplace. This leds to the assumption, that people try to consilidate their debts with cheaper loans on this side as descripted in the task introduciton.

3.5 Variable 4: `home_ownership`

3.5.1 Visualization

```
ggplot(myLCdata, aes(x = forcats::fct_infreq(home_ownership)))+
  geom_bar()+
  labs(
    title = "Home Ownership of Applicant",
    x = "Home Ownership",
    y = "Count"
  )
```

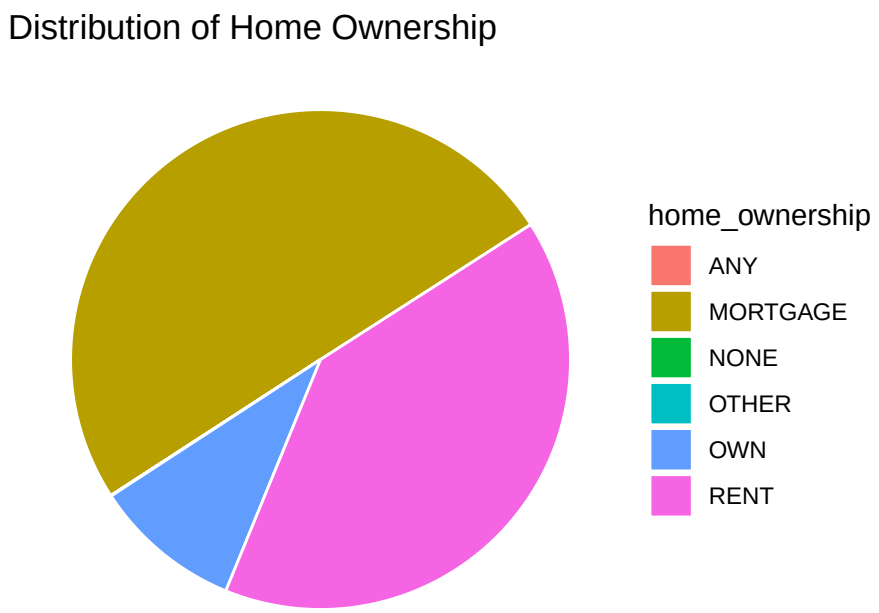


3.5.2 Interpretation

Only the categories mortgage, rent & own are relevant for this feature The other attributes are rarley used. Mortage and rent are used most often. People with home ownership probaly first get a loan by bank with a mortage. In most countries this is the cheapest way to get loans, because property serves as a security. This could be the reason the precentage of the ownership is so low. To best visualize the proportion we could also use a pie chart.

```
ggplot(myLCdata, aes(x="",fill = home_ownership))+
  geom_bar(width = 1, color = "white")+
  coord_polar("y", start=1)+
  labs(
    title = "Distribution of Home Ownership",
    file = "Home Ownership"
  )+
  theme_void()
```

```
Ignoring unknown labels:
* file : "Home Ownership"
```

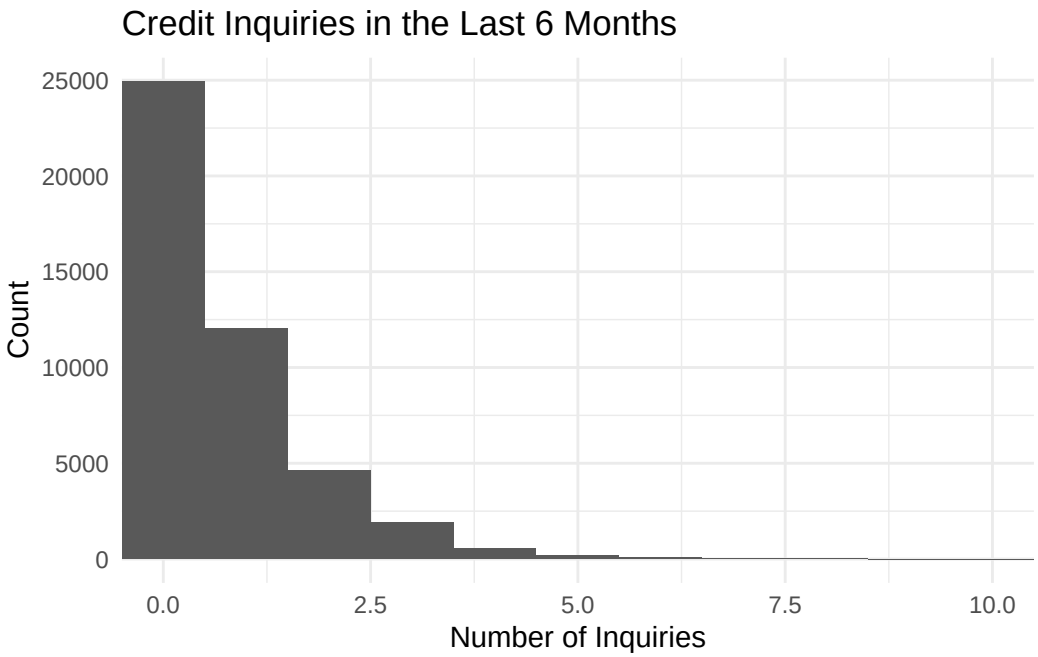


3.6 Variable 5: `inq_last_6mths`

3.6.1 Visualization

```
ggplot(myLCdata, aes(x = inq_last_6mths)) +
  geom_histogram(binwidth = 1) +
  coord_cartesian(xlim = c(0, 10)) +
  labs(
    title = "Credit Inquiries in the Last 6 Months",
    x = "Number of Inquiries",
    y = "Count"
  ) +
  theme_minimal()
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).
```



3.6.2 Interpretation

The distribution is highly right-skewed. Most borrowers have zero to a few inquiries, while higher values occur rarely. Therefor the histogram was

---

3.7 Overall Observations from Univariate EDA

- Which variables show skewness?
- Which variables show imbalance?
- Any surprising patterns?
- Do visual findings confirm summary statistics?

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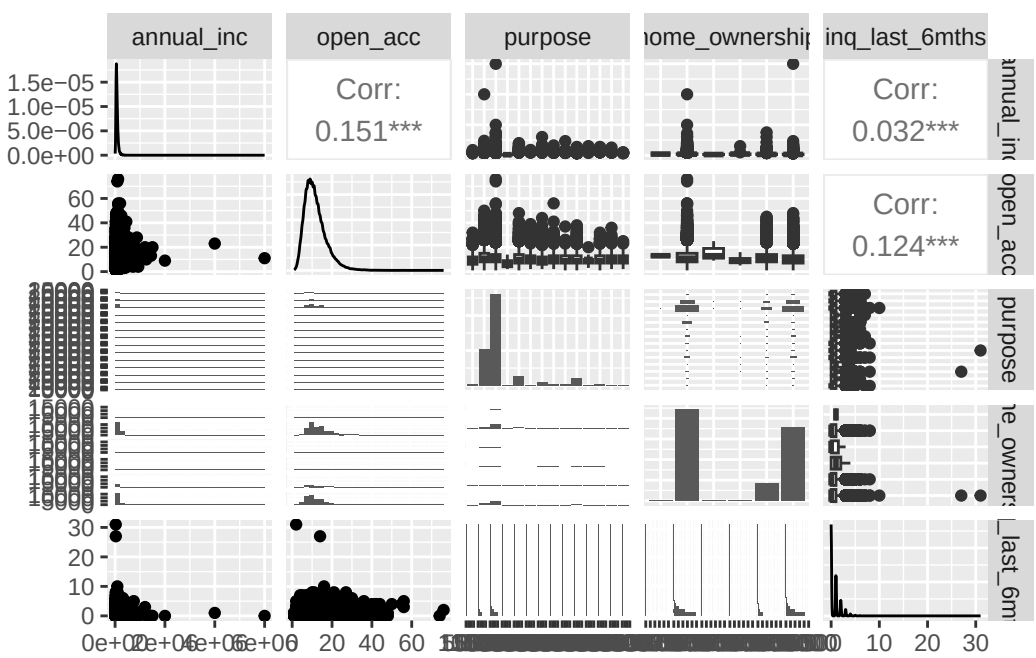
4 Task 4: Bivariate Exploratory Data Analysis (EDA)

4.1 Pairs Plot Overview

```r
if (!require("GGally")) install.packages("GGally")
library(GGally)

ggpairs(myLCdata)
```

```
Warning: Removed 2 rows containing missing values
Removed 2 rows containing missing values

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).
Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 2 rows containing missing values

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 2 rows containing non-finite outside the scale range (`stat_bin()`).
Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).
```



4.1.1 Initial Observations

- Are there visible correlations?
- Any clear patterns?
- Any surprising relationships?

4.2 Selected Insight 1

4.2.1 Variable Pair: _____ vs _____

4.2.2 Why This Pair Is Interesting

- What pattern was observed?
- Why was it not obvious before?

4.2.3 Stand-alone Professional Plot

```
# Recreate plot using ggplot2
```

4.2.4 Interpretation

- What does this relationship indicate?
- Business interpretation?

4.3 Selected Insight 2

4.3.1 Variable Pair: _____ vs _____

4.3.2 Stand-alone Professional Plot

```
# Recreate plot using ggplot2
```

4.3.3 Interpretation

Write here.

4.4 Adding a Third Variable

4.4.1 Enhanced Plot with Third Variable (e.g., color, shape, facet)

```
# ggplot with third aesthetic mapping
```

4.4.2 Why This Third Variable?

- Why is it relevant?
- What additional insight does it provide?
- What was not visible before?

4.5 Final Bivariate Insights

- Most important relationships discovered:
- Implications for credit risk assessment:
- Potential hypotheses for predictive modeling: